LESSON PLAN

HELICOPTER THEORY (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	11
Total No. of Lesson Plans	27	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Factors affecting CL, Theories of Lift production, force and pitching moment, coefficient of lift, variation of co-efficient of lift with aspect ratio and sweep back and CL
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Factors affecting CL, Theories of Lift production, force and pitching moment, coefficient of lift, variation of co-efficient of lift with aspect ratio and sweep back and CL.
5. Specific Objectives	11.1.2.1 Trainee will be able to describe various types of aerofoil sections, conditions of performance, theories, and concept related with performance

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of theoretical concepts via IB diagrams; Safety: Ensure no unauthorized handling of training materials; Supervision: Instructor monitors comprehension and directs discussion flow.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the four fundamental forces acting on an aircraft?
(b) Define the term 'angle of attack' in your own words.
(c) Recall the primary function of the wing in generating lift.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Types of aerofoil sections • Conditions of performance • Theories of lift production • Force and pitching moment concepts • Coefficient of lift definition	L & Ds	PPT, IB	15 min
(b)	• Variation of CL with aspect ratio • Effect of sweep back on CL • Relationship between CL and angle of attack • Critical angle of attack identification • Stall characteristics and recovery	L & Ds	PPT, PDF	20 min
(c)	• Summary of lift production theories • Application of CL concepts to helicopter rotors • Review of force and moment balance • Key takeaways on aerofoil performance • Preparation for pressure distribution study	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how increasing the aspect ratio affects the coefficient of lift.
(b) Describe the relationship between the angle of attack and the coefficient of lift up to the critical point.
(c) Differentiate between the center of pressure and the aerodynamic center based on today's lesson.

12. Home Assignment and Follow Up	Prepare a brief note comparing the lift generation mechanisms of a fixed-wing aircraft versus a helicopter rotor, referencing the coefficient of lift variations discussed in this lesson to bridge the gap to LP 12.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Study of pressure distribution over an aerofoil at various angles of attack, determination of center of pressure and aerodynamic center. various forces e.g. lift, drag and moments	Refer to the master syllabus reference list	LP 12

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Instructor, P No.	Senior Instructor, Senior Instructor, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min